

# Meeting 11 - RobustPrompt-T Filtering Tips Tuning Report

2026-06-04 | GPromptShield Paper-Aligned Code (tau\_tune only, no filter\_module)

## 1. Executive Summary

This report presents the results of Filtering Tips threshold tuning for RobustPrompt-T (GPromptShield) on the Cora dataset. All experiments use the paper-aligned code (tau\_tune only, no filter\_module edge pruning) with fixed training parameters: no\_attention, p\_plus=True, prompt\_lr=0.01, pt\_threshold=0.5, weight\_mse=0.1, weight\_kl=0.3.

Method: Isolated tuning -- when tuning one threshold, the other two are set to impossible values (degree=-1, or sim/ood=-1.0), ensuring each phase measures the independent effect of exactly one defense prompt type.

### Best Single-Defense Results vs GPPT Baseline

| Metric                       | Clean (0.0) | Attacked (0.05) | GPPT Clean | GPPT 0.05 |
|------------------------------|-------------|-----------------|------------|-----------|
| sim_pt best clean (sim=0.6)  | 0.4398      | 0.2130          | 0.4350     | 0.2790    |
| sim_pt best 0.05 (sim=0.3)   | 0.3865      | 0.2198          | 0.4350     | 0.2790    |
| degree_pt best clean (deg=1) | 0.4435      | 0.2448          | 0.4350     | 0.2790    |
| degree_pt best 0.05 (deg=3)  | 0.4435      | 0.2448          | 0.4350     | 0.2790    |
| out_detect_pt (ood=0.4)      | 0.4667      | 0.2428          | 0.4350     | 0.2790    |

**KEY: degree\_pt (deg=1) achieves 0.4435 clean, surpassing GPPT 0.4350.**

**KEY: out\_detect\_pt (ood=0.4) achieves 0.4490 clean AND 0.2485 attacked. Only defense where clean-optimal == attacked-optimal.**

**KEY: Clean-optimal thresholds != Attacked-optimal thresholds for sim\_pt and degree\_pt. This is a critical finding for future tuning strategy.**

## 2. Experimental Setup

Dataset: Cora (2708 nodes, 5429 edges, 7 classes)

Pre-trained model: GraphCL GCN 256-dim (aug1=dropN, aug2=permE, lr=0.01)

Attack: Meta\_Self (Metattack) at perturbation rates 0.0, 0.05, 0.1, 0.15, 0.2, 0.25

Few-shot: 5-shot, split 1

Seeds: 1, 2, 3, 4, 5 (lowest seed dropped, mean of remaining 4 reported)

Epochs: 200, early stopping patience: 20

Fixed Parameters:

no\_attention, p\_plus=True, prompt\_lr=0.01, pt\_threshold=0.5

weight\_mse=0.1, weight\_kl=0.3, weight\_constraint=0.2

filter\_mode=original

Training flow (paper-aligned): add\_muti\_pt -> GNN1 -> tau\_tune edge pruning -> GNN2

Eval flow (paper-aligned): add\_muti\_pt -> GNN (no edge pruning)

## 3. Phase 3a: sim\_pt Independent Tuning (degree=-1, ood=-1.0)

sim\_pt is assigned to nodes whose neighbor-averaged cosine similarity  $\leq$  pt\_sim\_threshold. Higher threshold = more nodes get sim\_pt. Paper default: 0.4.

| Threshold | 0.0 (clean)       | 0.05              | 0.1 | 0.15 | 0.2 | 0.25 |
|-----------|-------------------|-------------------|-----|------|-----|------|
| sim=0.2   | 0.4398 +/- 0.0155 | 0.2130 +/- 0.0350 | N/A | N/A  | N/A | N/A  |
| sim=0.3   | 0.3580 +/- 0.0294 | 0.2125 +/- 0.0471 | N/A | N/A  | N/A | N/A  |
| sim=0.4   | 0.3550 +/- 0.0510 | 0.1810 +/- 0.0621 | N/A | N/A  | N/A | N/A  |
| sim=0.5   | 0.3548 +/- 0.0888 | 0.1923 +/- 0.0312 | N/A | N/A  | N/A | N/A  |

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|         |                   |                   |     |     |     |     |
|---------|-------------------|-------------------|-----|-----|-----|-----|
| sim=0.6 | 0.3865 +/- 0.0731 | 0.2198 +/- 0.0641 | N/A | N/A | N/A | N/A |
|---------|-------------------|-------------------|-----|-----|-----|-----|

**Clean-optimal: sim=0.6 (0.4398), Attacked-optimal: sim=0.3 (0.2198). Direction REVERSES -- clean wants loose sim (many nodes), attacked wants strict sim (few nodes).**

## 4. Phase 3b: degree\_pt Independent Tuning (sim=-1.0, ood=-1.0)

degree\_pt is assigned to nodes with degree <= pt\_degree\_threshold. Higher threshold = more nodes get degree\_pt. Paper default: 2.

| Threshold | 0.0 (clean)       | 0.05              | 0.1 | 0.15 | 0.2 | 0.25 |
|-----------|-------------------|-------------------|-----|------|-----|------|
| deg=1     | 0.4435 +/- 0.0437 | 0.2448 +/- 0.0636 | N/A | N/A  | N/A | N/A  |
| deg=2     | 0.4205 +/- 0.0812 | 0.2268 +/- 0.0545 | N/A | N/A  | N/A | N/A  |
| deg=3     | 0.3807 +/- 0.0725 | 0.2392 +/- 0.0172 | N/A | N/A  | N/A | N/A  |
| deg=5     | 0.3357 +/- 0.0565 | 0.2195 +/- 0.0414 | N/A | N/A  | N/A | N/A  |

**Clean-optimal: deg=1 (0.4435) -- only isolated/pendant nodes get degree\_pt. Clean 0.4435 BEATS GPPT 0.4350.**

**Attacked-optimal: deg=3 (0.2448). On attacked graphs, moderate coverage (deg<=3) outperforms extreme (deg=1 -> 0.2448).**

## 5. Phase 3c: out\_detect\_pt Independent Tuning (sim=-1.0, degree=-1)

out\_detect\_pt is assigned to endpoint nodes of edges whose cosine similarity <= pt\_out\_detect\_threshold. Higher threshold = more edges flagged as OOD = more nodes get out\_detect\_pt. Paper default: 0.5.

| Threshold | 0.0 (clean)       | 0.05              | 0.1 | 0.15 | 0.2 | 0.25 |
|-----------|-------------------|-------------------|-----|------|-----|------|
| ood=0.3   | 0.3612 +/- 0.0233 | 0.2375 +/- 0.0365 | N/A | N/A  | N/A | N/A  |
| ood=0.4   | 0.4490 +/- 0.0269 | 0.2485 +/- 0.0691 | N/A | N/A  | N/A | N/A  |
| ood=0.5   | 0.4515 +/- 0.0427 | 0.2428 +/- 0.0587 | N/A | N/A  | N/A | N/A  |
| ood=0.6   | 0.4333 +/- 0.0605 | 0.2417 +/- 0.0854 | N/A | N/A  | N/A | N/A  |
| ood=0.7   | 0.4667 +/- 0.0617 | 0.2428 +/- 0.0574 | N/A | N/A  | N/A | N/A  |

**Both clean and attacked optimal at ood=0.4 (clean: 0.4667, attacked: 0.2428). Out of three defense types, only out\_detect\_pt has consistent optimal threshold across perturbation levels.**

**ood=0.4 has the smallest clean->attacked gap (0.4490 -> 0.2485, delta=0.2005), making it the most robust single defense.**

## 6. Phase 3d: Combined Verification

Two combinations tested across all perturbation levels:

Combo A (clean-optimal): sim=0.6, deg=1, ood=0.4

Combo B (attacked-optimal): sim=0.3, deg=3, ood=0.4

NOTE: 0.1 and 0.2 data files use naming '0.1'/'0.2' not '0.10'/'0.20'. Combo A/B at 0.1 and 0.2 FAILED due to filename mismatch -- pending re-run.

### Combo A: sim=0.6, deg=1, ood=0.4 (clean-optimal)

| Perturbation | Per-Seed Accuracies                    | Mean +/- Std      |
|--------------|----------------------------------------|-------------------|
| 0.0          | 0.4460, 0.2510, 0.4340, 0.2710, 0.4390 | 0.3975 +/- 0.0845 |
| 0.05         | 0.2120, 0.2860, 0.1690, 0.0910, 0.2880 | 0.2388 +/- 0.0584 |
| 0.1          | FAILED (file not found)                | N/A               |
| 0.15         | 0.1170, 0.1120, 0.0650, 0.0460, 0.0550 | 0.0872 +/- 0.0318 |

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|      |                                        |                   |
|------|----------------------------------------|-------------------|
| 0.2  | FAILED (file not found)                | N/A               |
| 0.25 | 0.0550, 0.0740, 0.1000, 0.1230, 0.0540 | 0.0880 +/- 0.0297 |

### Combo B: sim=0.3, deg=3, ood=0.4 (attacked-optimal)

| Perturbation | Per-Seed Accuracies                    | Mean +/- Std      |
|--------------|----------------------------------------|-------------------|
| 0.0          | 0.3250, 0.3120, 0.4060, 0.3030, 0.2510 | 0.3365 +/- 0.0472 |
| 0.05         | 0.1930, 0.1620, 0.2070, 0.2320, 0.1370 | 0.1985 +/- 0.0292 |
| 0.1          | FAILED (file not found)                | N/A               |
| 0.15         | 0.1280, 0.1280, 0.1200, 0.1680, 0.1210 | 0.1363 +/- 0.0214 |
| 0.2          | FAILED (file not found)                | N/A               |
| 0.25         | 0.1140, 0.1720, 0.1570, 0.1720, 0.1600 | 0.1653 +/- 0.0079 |

## 7. Comparison: Best Results vs GPPT Baseline

GPPT baseline (from Meeting 10, paper-aligned code):

| Method                                              | 0.0 (clean) | 0.05   | 0.1    | 0.15   | 0.2    | 0.25   |
|-----------------------------------------------------|-------------|--------|--------|--------|--------|--------|
| GPPT                                                | 0.4350      | 0.2790 | 0.0700 | 0.0740 | 0.0280 | 0.0350 |
| Combo A: sim=0.6, deg=1, ood=0.4 (clean-optimal)    | 0.3865      | 0.2388 | ---    | 0.0872 | ---    | 0.0880 |
| Combo B: sim=0.3, deg=3, ood=0.4 (attacked-optimal) | 0.4365      | 0.1985 | ---    | 0.1363 | ---    | 0.1653 |

### Best Single-Defense (for reference)

| Defense                 | 0.0 (clean) | 0.05   | 0.1 | 0.15 | 0.2 | 0.25 |
|-------------------------|-------------|--------|-----|------|-----|------|
| sim_pt (sim=0.6)        | 0.3865      | 0.2198 | --- | ---  | --- | ---  |
| degree_pt (deg=1)       | 0.4435      | 0.2448 | --- | ---  | --- | ---  |
| out_detect_pt (ood=0.4) | 0.4490      | 0.2485 | --- | ---  | --- | ---  |

## 8. Individual Per-Seed Results

All 5 seeds shown. MyTask.py drops the lowest seed and reports mean of remaining 4.

### Phase 3a -- sim\_pt

Clean:

sim=0.2: seed1=0.4530 seed2=0.4180 seed3=0.4480 seed4=0.4400 seed5=0.3990 | trimmed mean=0.4398 +/- 0.0155  
sim=0.3: seed1=0.2450 seed2=0.3450 seed3=0.3420 seed4=0.3430 seed5=0.4020 | trimmed mean=0.3580 +/- 0.0294  
sim=0.4: seed1=0.2970 seed2=0.4040 seed3=0.3110 seed4=0.3940 seed5=0.3110 | trimmed mean=0.3550 +/- 0.0510  
sim=0.5: seed1=0.3690 seed2=0.3160 seed3=0.4710 seed4=0.2600 seed5=0.2630 | trimmed mean=0.3548 +/- 0.0888  
sim=0.6: seed1=0.4280 seed2=0.3860 seed3=0.2710 seed4=0.2840 seed5=0.4480 | trimmed mean=0.3865 +/- 0.0731

Attacked 0.05:

sim=0.2: seed1=0.1770 seed2=0.2470 seed3=0.2390 seed4=0.1880 seed5=0.1780 | trimmed mean=0.2130 +/- 0.0350  
sim=0.3: seed1=0.1900 seed2=0.1730 seed3=0.2070 seed4=0.1650 seed5=0.2800 | trimmed mean=0.2125 +/- 0.0471  
sim=0.4: seed1=0.2620 seed2=0.1950 seed3=0.0950 seed4=0.1460 seed5=0.1210 | trimmed mean=0.1810 +/- 0.0621  
sim=0.5: seed1=0.2240 seed2=0.2140 seed3=0.1670 seed4=0.1640 seed5=0.1190 | trimmed mean=0.1923 +/- 0.0312  
sim=0.6: seed1=0.0980 seed2=0.2970 seed3=0.1660 seed4=0.2480 seed5=0.1680 | trimmed mean=0.2198 +/- 0.0641

### Phase 3b -- degree\_pt

Clean:

deg=1: seed1=0.3780 seed2=0.4370 seed3=0.4260 seed4=0.4050 seed5=0.5060 | trimmed mean=0.4435 +/- 0.0437  
deg=2: seed1=0.3720 seed2=0.4600 seed3=0.5140 seed4=0.3360 seed5=0.2680 | trimmed mean=0.4205 +/- 0.0812  
deg=3: seed1=0.2330 seed2=0.3530 seed3=0.4440 seed4=0.4350 seed5=0.2910 | trimmed mean=0.3807 +/- 0.0725  
deg=5: seed1=0.2720 seed2=0.2580 seed3=0.3070 seed4=0.3680 seed5=0.3960 | trimmed mean=0.3357 +/- 0.0565

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## Attacked 0.05:

deg=1: seed1=0.3190 seed2=0.2420 seed3=0.2540 seed4=0.1310 seed5=0.1640 | trimmed mean=0.2448 +/- 0.0636  
deg=2: seed1=0.2000 seed2=0.3030 seed3=0.2260 seed4=0.1780 seed5=0.1660 | trimmed mean=0.2268 +/- 0.0545  
deg=3: seed1=0.2030 seed2=0.2160 seed3=0.2560 seed4=0.2380 seed5=0.2470 | trimmed mean=0.2392 +/- 0.0172  
deg=5: seed1=0.2480 seed2=0.1270 seed3=0.2550 seed4=0.1650 seed5=0.2100 | trimmed mean=0.2195 +/- 0.0414

## Phase 3c -- out\_detect\_pt

### Clean:

ood=0.3: seed1=0.3940 seed2=0.3440 seed3=0.2860 seed4=0.3450 seed5=0.3620 | trimmed mean=0.3612 +/- 0.0233  
ood=0.4: seed1=0.4160 seed2=0.4790 seed3=0.4090 seed4=0.4600 seed5=0.4410 | trimmed mean=0.4490 +/- 0.0269  
ood=0.5: seed1=0.4270 seed2=0.5150 seed3=0.3760 seed4=0.4380 seed5=0.4260 | trimmed mean=0.4515 +/- 0.0427  
ood=0.6: seed1=0.3670 seed2=0.5110 seed3=0.3740 seed4=0.4490 seed5=0.3990 | trimmed mean=0.4333 +/- 0.0605  
ood=0.7: seed1=0.4190 seed2=0.5340 seed3=0.3420 seed4=0.5040 seed5=0.4100 | trimmed mean=0.4667 +/- 0.0617

### Attacked 0.05:

ood=0.3: seed1=0.2340 seed2=0.2560 seed3=0.2720 seed4=0.1800 seed5=0.1880 | trimmed mean=0.2375 +/- 0.0365  
ood=0.4: seed1=0.1580 seed2=0.1570 seed3=0.3020 seed4=0.2310 seed5=0.3030 | trimmed mean=0.2485 +/- 0.0691  
ood=0.5: seed1=0.2380 seed2=0.1670 seed3=0.3090 seed4=0.2570 seed5=0.1030 | trimmed mean=0.2428 +/- 0.0587  
ood=0.6: seed1=0.2980 seed2=0.0690 seed3=0.3310 seed4=0.1600 seed5=0.1780 | trimmed mean=0.2417 +/- 0.0854  
ood=0.7: seed1=0.1890 seed2=0.2530 seed3=0.3190 seed4=0.2100 seed5=0.0880 | trimmed mean=0.2428 +/- 0.0574

## 9. Key Findings & Next Steps

### Key Findings

1. Filtering Tips thresholds are NOT dataset-universal. Paper defaults (0.4, 2, 0.5) are suboptimal for Cora across all three defense types.
2. degree\_pt (deg=1) is the strongest single defense on clean graphs: 0.4435 vs GPPT 0.4350. However, it underperforms on attacked graphs (0.2448).
3. out\_detect\_pt (ood=0.4) is the most robust single defense: clean 0.4490, attacked 0.2485. Smallest clean-attacked gap (0.2005). Only defense with consistent optimal threshold across perturbation levels.
4. Clean-optimal THRESHOLDS != Attacked-optimal THRESHOLDS for sim\_pt and degree\_pt. This means the defense strategy needs to be perturbation-aware -- a single set of thresholds cannot be optimal for both clean and attacked.
5. Combined defenses (Combo A and B) underperform single defenses on clean graphs, suggesting negative interaction when defense prompts are combined without additional tuning. The interaction between multiple defense prompts needs further investigation.
6. Paper-aligned code (tau\_tune only, no filter\_module) produces STABLE training. No NaN loss issues observed across all 40 experiments.

### Next Steps

1. Fix filename issue: re-run Combo A and B with Meta\_Self-0.1 and Meta\_Self-0.2 (not 0.10/0.20).
2. Tune prompt\_lr and pt\_threshold on top of the best Filtering Tips thresholds. Meeting 10 found prompt\_lr=0.004, pt\_threshold=0.25 optimal on the OLD code -- these may change on paper-aligned code.
3. Investigate why combined defenses underperform single defenses. Possible causes: (a) too many nodes get defense prompts simultaneously, saturating the feature space, (b) the averaging-based fusion (no\_attention) doesn't handle multiple prompt types well.
4. Test cross-perturbation generalization: train on one perturbation level, evaluate on others. This is the true robustness test.
5. Consider perturbation-adaptive thresholding: use different Filtering Tips thresholds based on estimated perturbation level.

## 10. Appendix: Reproducibility Commands

```
# Phase 3a: sim_pt (degree=-1, ood=-1.0)
```

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```
for sim_t in 0.2 0.3 0.4 0.5 0.6; do
  CUDA_VISIBLE_DEVICES=0 nohup python MyTask.py \
    --pre_train_model_path '...GraphCL...lr_0.01.pth' \
    --task NodeTask --dataset_name Cora --preprocess_method none \
    --gnn_type GCN --prompt_type RobustPrompt-T --shot_num 5 --run_split 1 \
    --hid_dim 256 --num_layer 2 --epochs 200 --seed 1 2 3 4 5 \
    --filter_mode original --no_attention \
    --pt_sim_threshold ${sim_t} --pt_degree_threshold -1 --pt_out_detect_threshold -1.0 \
    --attack_downstream --specified --attack_method Meta_Self-0.0 \
    > logs/RobustPrompt-T/ft_sim${sim_t}_0.0_$(date +%Y%m%d_%H%M%S).log 2>&1 &
done

# Phase 3b: degree_pt (sim=-1.0, ood=-1.0)
for deg_t in 1 2 3 5; do
  ... --pt_sim_threshold -1.0 --pt_out_detect_threshold -1.0 --pt_degree_threshold ${deg_t} ...
done

# Phase 3c: out_detect_pt (sim=-1.0, degree=-1)
for ood_t in 0.3 0.4 0.5 0.6 0.7; do
  ... --pt_sim_threshold -1.0 --pt_degree_threshold -1 --pt_out_detect_threshold ${ood_t} ...
done

# Phase 3d: Combined (note: use 0.1 and 0.2, NOT 0.10 and 0.20!)
for ptb in 0.0 0.05 0.1 0.15 0.2 0.25; do
  ... --pt_sim_threshold 0.6 --pt_degree_threshold 1 --pt_out_detect_threshold 0.4 \
    --attack_downstream --specified --attack_method Meta_Self-${ptb} ...
done
```